# Viva Script — FE: What We Did to the Features

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## 1. Opening (about 30 seconds)

Read the script as written, or use the bold lines as prompts. Lines in *[brackets]* are cues for what to show.

> "One operating cycle of our hydraulic rig lasts 60 seconds and produces **43,680 raw numbers** from 17 sensors. We have only **2,205 cycles**. You cannot hand 43,680 columns to a model that has 2,205 rows to learn from.
>
> There are two more problems. The sensors run at 100, 10 and 1 hertz, so sample number k of a pressure signal and sample number k of a temperature signal are not even the same moment in time. And the faults we want to detect show up as **relationships between sensors**, which no single raw sample shows.
>
> Feature engineering solves all three. In this part I will explain what features we created, how we combined sensors, what we removed, how we selected the final features, and what the models ended up using."

*[Show FE deck, slide 2]*

## 2. Creating features: five blocks, 626 features

> "We treat each 60-second cycle as one row. From every sensor we calculate numbers that describe the shape of its signal, so a cycle of 43,680 raw values becomes 626 named features, about 70 times smaller. Each name tells you exactly what it is, for example PS2_cv is the coefficient of variation of pressure sensor 2."

| Block | Count | What it measures | Real example from our data |
|---|---|---|---|
| Statistical | 251 | The distribution of each signal over the cycle: mean, median, std, min, max, RMS, skewness, kurtosis, quartiles, coefficient of variation | PS2_cv, FS1_q75 (75th percentile of flow) |

| Block | Count | What it measures | Real example from our data |
|---|---|---|---|
| Time-domain | 162 | How the signal changes: slope, first-difference statistics, zero crossings, number of peaks | PS2_max_abs_change, PS2_std_diff |
| Rolling | 54 | Local behaviour in a 1-second moving window (fast sensors only) | PS3_rollmean_max, PS1_rollstd_max |
| Frequency | 63 | FFT summary: dominant frequency, spectral centroid, energy in 3 bands (fast sensors only) | PS1_band3_energy_ratio |
| Hydraulic domain | 96 | Physical relationships between sensors (next section) | dp_PS1_PS2_mean, efficiency_PS1_FS1 |

Two design choices worth saying out loud:

1. **The rolling window is one second for every sensor**, converted using each sensor's own rate: 100 samples at 100 Hz, 10 samples at 10 Hz. A fixed sample count would mean 1 second for one sensor and 100 seconds for another.
2. **The frequency bands are log-spaced**, from 1/60 Hz up to the Nyquist limit. The load changes every ~10 seconds, so almost all energy sits below 1 Hz. With equal-width bands, the first band would hold nearly everything and the others would be constant zeros.

> "Why statistics and not raw samples? Faults change the shape of a signal before they change its average. A valve with switching lag spends longer in transition, which widens the pressure distribution even when mean pressure is unchanged. Standard deviation, skewness and kurtosis capture that."

## 3. Combining sensors: the hydraulic physics features

> "This is where we combine sensors. Instead of letting a model guess how channels relate, we wrote the relationships down using hydraulic physics. That gave 96 domain features, each built from two or more sensors."

| Combination | How it is built | Fault it targets |
|---|---|---|
| Pressure differentials and ratios | mean of one pressure sensor minus, or divided by, another, for all 15 pairs of PS1–PS6 | Valve lag, internal leakage |
| Flow imbalance | FS1 minus FS2, their ratio and relative difference | Pump leakage |
| Specific power | motor power EPS1 divided by flow | Pump leakage |
| Hydraulic power | pressure times flow / 600, in kW | Pump, valve |
| Electro-hydraulic efficiency | hydraulic power / electrical power | Pump wear |
| Hydraulic resistance | pressure / flow | Valve, accumulator |
| Temperature spreads | differences between TS1–TS4, spread, maximum rise | Cooler |
| Cooling conductance | cooling power CP / temperature difference, CP / cooling efficiency CE | Cooler |
| Vibration per unit power | VS1 / EPS1 | Mechanical wear |

**Formulas:** hydraulic power $P_h$ [kW] = p [bar] × Q [l/min] / 600, and efficiency $\eta = 1000 \times P_h$ / EPS1 [W].

> "Here is why this matters. A leaking pump still draws roughly the same electrical power, but it delivers less flow. Each sensor on its own can look normal, but efficiency drops. Our efficiency feature makes that visible in a single number."

If asked whether we used PCA or other automatic combination: **no.** We combined sensors only through physics, so every feature stays interpretable. PCA would mix hundreds of features into components nobody can name, so we used it only for visualisation plots, never as model input.

## 4. Cleaning: 626 to 553 features

> "Before any model sees the features, we remove columns that cannot help. These checks never look at the labels, so running them on all cycles cannot leak any information about the answer."

| Removed | Count | Why | Example |
|---|---|---|---|
| Constant | 4 | The same value in every cycle carries no information | PS2_min |
| Near-zero variance | 4 | Almost constant, so effectively noise | PS5_slope |
| Exact duplicates | 65 | Identical to another column | PS1_peak_to_peak is identical to PS1_range |

> "The duplicates happen naturally: peak-to-peak and range are two names for maximum minus minimum. After cleaning we have **553 features**."

Inside every model pipeline, two more fixed steps prepare the features:

- **Median imputation:** any missing value is filled with the training median (robust to outliers in ratio features).
- **Standard scaling:** each feature becomes a z-score, using the training mean and standard deviation, so no feature dominates just because of its units.

## 5. Selection: 553 to 40 features

> "553 features is still a lot, and many say nearly the same thing. So for the 'selected' feature set we run three filters in a row. Each one is a different way of asking 'is this feature useful?'"

| Stage | What it does | Valve model | Pump model |
|---|---|---|---|
| Start | Cleaned features | 553 | 553 |
| 1. Correlation filter | Of any two features with correlation above 0.95, keeps the first and drops the other. Removes redundancy. | 238 | 236 |
| 2. Mutual information | Keeps the top 50 % by mutual information with the label, a model-free score that also catches non-linear links | 119 | 118 |

| Stage | What it does | Valve model | Pump model |
|---|---|---|---|
| 3. Tree importance | A 200-tree random forest ranks the survivors together, and the top 40 are kept | **40** | **40** |

> "The three stages complement each other. The correlation filter removes duplicates. Mutual information judges each feature alone against the label. The tree stage judges features in the context of each other, so it can keep a feature that only helps in combination."

**The most important point: selection happens inside the pipeline.**

> "Selection uses the labels. If we had selected features once on the whole dataset, the test cycles would already have influenced which features were chosen, and our scores would be inflated. So selection runs inside the scikit-learn Pipeline and is re-fitted on every cross-validation training fold. That is what 'leakage-safe' means."

Which blocks the 40 selected features came from:

| Model | Statistical | Time-domain | Rolling | Frequency | Domain |
|---|---|---|---|---|---|
| Valve | 9 | 11 | 5 | 8 | 7 |
| Pump | 20 | 6 | 3 | 2 | 9 |

> "The valve model leans on time-domain and frequency features, which describe how sharply pressure switches. The pump model leans on statistics of flow and motor power, plus domain features. Every block survived into at least one model."

## 6. What each model uses, and what feature engineering bought

> "We tested three feature sets on every model: a baseline of just the 17 sensor means, all 553 engineered features, and the 40 selected features. The winner for each component used a different set."

| Component | Feature set that won | Inputs | Most important feature (test set) | Physical meaning |
|---|---|---|---|---|
| Cooler | Baseline | 17 sensor means | CP_mean | Cooling power directly measures heat removal |
| Valve | Selected | 40 | PS2_cv | Switching lag changes how much pressure varies |
| Pump leakage | Selected | 40 | FS1_q75 | Leakage cuts delivered flow |
| Accumulator | Engineered (no selection) | 553 | spread thinly over many features | Correlated rolling and frequency features of PS1 and PS3 |

*[Show FE deck, slide 6]*

> "The clearest proof of what feature engineering bought is logistic regression, because a linear model cannot build features by itself. On the valve it went from **0.672 to 0.991** macro-F1. On the accumulator, **0.642 to 0.977**. The SVM on the valve went from 0.529 to 0.989."

| Component | Best with 17 sensor means | Best with engineered features | Gain |
|---|---|---|---|
| Cooler | 0.998 | 0.998 | none |
| Valve | 0.968 | 0.995 | +0.026 |
| Pump leakage | 0.991 | 0.995 | +0.005 |
| Accumulator | 0.971 | 0.992 | +0.022 |

> "For the strongest models the gain is smaller, 2 to 3 points on the valve and accumulator, because tree ensembles build non-linear combinations internally. And for the cooler, engineering added nothing, because the sensor means already separate its three levels. We say that honestly rather than hide it."

## 7. Advantages and drawbacks of our feature engineering

### Advantages

- **Solves the size problem:** 43,680 raw values become 626 features, and just 40 for the valve and pump models.
- **Fixes the sampling-rate mismatch:** every feature summarises a whole cycle, so sensors at 100, 10 and 1 Hz become comparable.
- **Encodes physics:** domain features such as efficiency and pressure differentials express exactly what a fault changes.
- **Interpretable:** every feature has a readable name, so an engineer can check an alarm against the physics.
- **Makes simple models strong:** logistic regression on the valve jumps from 0.672 to 0.991.
- **Leakage-safe:** cleaning uses no labels, and selection is re-fitted inside each training fold.
- **Fast and reusable:** the same code extracts features from a new cycle in well under a second.

**Drawbacks and where it falls short**

1. **Summaries lose timing detail.** A whole-cycle statistic cannot say *when* in the cycle something happened. Transients right after each load step, which may separate the two lowest accumulator levels, are averaged away.
2. **Needs domain knowledge.** The physics features required understanding hydraulics. A new machine type would need new features.
3. **Many correlated features.** 65 exact duplicates had to be removed, and the correlation filter still drops more than half of the rest. The accumulator model, with no selection, keeps all 553 correlated features, which makes it hard to explain.
4. **No gain on easy targets.** The cooler scored 0.998 with just the 17 sensor means. Engineering added nothing there.
5. **Relies on virtual channels.** Features built on CE, CP and SE inherit the rig's own calculations.
6. **Fixed recipe.** Window length, number of bands and the 0.95 correlation threshold were chosen by reasoning, not tuned.

> "So feature engineering was essential for the valve and accumulator, and for making simple models competitive. Its main limitation is that summarising a whole cycle hides short events inside it."

# 8. Why the scores are so high, and why that is not 'perfect'

> "Our best models reach 0.99 to 1.00. Part of that is our features, and part of it is the dataset itself. It is important to separate the two."

**What the features contribute**

- They make the classes nearly **linearly separable**. That is why even logistic regression reaches 0.99 on the valve once it has engineered features.
- The physics features line up directly with the faults: cooling power with the cooler, flow with pump leakage, pressure variation with the valve.

**What the data contributes**

1. **A controlled lab rig:** faults set on purpose at a few well-separated levels, under the same 60-second load cycle every time.
2. **Conditions held for long runs:** in recording order the cooler changes level only 3 times (about 732 cycles per level), the accumulator 12 times, the pump 37 times.
3. **Near-duplicate neighbours:** consecutive cycles are almost identical, so with a random split nearly every test cycle has a twin in training. Blocked validation, which holds out whole stretches of time, drops the accumulator from 0.992 to 0.916.

**Is it a perfect result?** No.

> "A perfect classifier would be right on every future cycle, from any machine. We cannot prove that. Zero errors on 441 test cycles still allows a true error rate of up to about 0.7 % at 95 % confidence, the rule of three: 3 divided by 441. The honest figure is the blocked validation, 0.92 to 0.995. Even the best possible classifier, the Bayes-optimal one, makes errors wherever two classes truly overlap, as our 90 and 100 bar accumulator cycles do."

> "So our claim is modest: the features make the fault levels of this rig easy to separate. Proving it on real plant data is the next step."

# 9. Closing and likely viva questions

> "To summarise: we turned 43,680 raw values per cycle into 626 named features in five blocks, including 96 physics features that combine sensors. We cleaned them to 553, and a leakage-safe three-stage filter selected 40 for the valve and pump. These features made even a linear model near-perfect on the valve, and every important feature has a physical explanation. Thank you."

| Question | Answer in one or two lines |
|---|---|
| Why not feed raw samples to a deep network? | 43,680 inputs against 2,205 cycles would overfit, the sensors are not time-aligned, and the result would not be interpretable. |
| What is the difference between feature extraction and feature selection? | Extraction creates new features from raw signals (626). Selection chooses a useful subset of them (40). |
| Did you combine features automatically, for example with PCA? | No. We combined sensors through physics (differentials, efficiency, conductance) to stay interpretable. PCA was only used for plots. |
| Why three selection stages instead of one? | Each catches something different: duplicates (correlation), single-feature relevance (mutual information), usefulness in combination (tree importance). |
| Why must selection be inside the pipeline? | It uses labels. Selecting on all data first would leak test information and inflate scores. |
| Is cleaning on all data not leakage too? | No. Constant, near-constant and duplicate checks never look at the labels. |
| Why is the rolling window one second? | Long enough to smooth noise, short enough to keep local behaviour. It is converted per sensor rate so it means the same time everywhere. |
| Why log-spaced frequency bands? | Almost all signal energy is below 1 Hz. Equal-width bands would leave most bands empty. |
| Why does the accumulator use all 553 features? | That configuration scored highest in cross-validation. Adding selection to it is listed as future work. |

| Question | Answer in one or two lines |
|---|---|
| Why did engineering not help the cooler? | Its levels are already separated by the cooling channels' means, so 17 sensor means reach 0.998. |
| What would you add next? | Features around each load step to capture transients, and validation on data from other machines. |